

GYNECOLOGY

Intracervical block for levonorgestrel-releasing intrauterine system placement among nulligravid women: a randomized double-blind controlled trial



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BACKGROUND: Fear of pain during the insertion of intrauterine contraceptives is a barrier to using these methods, especially for nulligravidas. An intracervical block may be easier and more reproducible than a paracervical block; however, this intervention has not been evaluated in nulligravid women to reduce pain with intrauterine contraceptive insertion.

OBJECTIVE: To evaluate whether a 3.6-mL 2% lidocaine intracervical block reduces pain at tenaculum placement and levonorgestrel-releasing intrauterine system insertion among nulligravidas; and, in addition, to assess whether the intracervical block has any effect on the ease of device insertion and on the overall experience with the procedure.

MATERIALS AND METHODS: In this randomized double-blind controlled trial, nulligravidas were block-randomized to 1 of 3 arms prior to 52-mg levonorgestrel-releasing intrauterine system insertion: 3.6-mL 2%-lidocaine intracervical block, sham injection (intracervical dry-needling), or no intervention. The primary outcome was pain at levonorgestrel-releasing intrauterine system insertion. Secondary outcomes were pain at tenaculum placement, ease of insertion (assessed by healthcare providers), and the overall experience with the procedure (pain with levonorgestrel-releasing intrauterine system insertion compared with expectations, discomfort level, wish to undergo another device insertion in the future, and recommendation of the procedure to others). Participants' pain was measured with a 10-cm visual analogue scale and a 5-point Faces Pain Scale. Pain was summarized into categories (none, mild, moderate, severe) and also analyzed as a continuous variable (mean and 95% confidence interval). Our sample size had 80% power ($\alpha = 0.05$) to detect a 15% difference in pain score measured by visual analogue scale (mean [standard deviation] visual analogue scale score = 5.9 [2.0] cm) and an absolute difference of 20% in the proportion of women reporting severe pain at levonorgestrel-releasing intrauterine system insertion among groups. We used a χ^2 test and a mixed-effects linear regression

model. We calculated the number needed to treat for the intracervical block to avert severe pain at tenaculum placement and levonorgestrel-releasing intrauterine system insertion.

RESULTS: A total of 302 women were randomized (99 to the intracervical block, 101 to the intracervical sham, and 102 to no intervention), and 300 had a successful device insertion. The intracervical block group had fewer women reporting severe pain than the other groups, both at tenaculum placement (intracervical block: 2% vs sham: 30.2% vs no intervention: 15.2%, $P < .0001$) and at levonorgestrel-releasing intrauterine system insertion (intracervical block: 26.5% vs sham: 59.4% vs no intervention: 50.5%, $P < .0001$). The mean (95% confidence interval) pain score reported at levonorgestrel-releasing intrauterine system insertion was lower in the intracervical block group than in the other groups (intracervical block: 4.3 [3.8–4.9] vs sham: 6.6 [6.2–7.0], $P < .0001$; intracervical block: 4.3 [3.8–4.9] vs no intervention: 5.8 [5.3–6.4], $P < .0001$). Women from the intracervical block group reported less pain than expected ($P < .0001$), rated the insertion as less uncomfortable ($P < .0001$), and were more willing to undergo another device insertion in the future ($P < .01$) than women in the other groups. The ease of insertion were similar among groups. The number needed to treat for the intracervical block to avert severe pain at tenaculum placement and levonorgestrel-releasing intrauterine system insertion was 2 and 4, respectively.

CONCLUSION: A 3.6-mL 2% lidocaine intracervical block decreased pain at tenaculum placement and levonorgestrel-releasing intrauterine system insertion among nulligravidas. It also provided a better overall experience during the procedure.

Key words: insertion, intracervical block, levonorgestrel-releasing intrauterine system, lidocaine, nulligravid women, pain

Intrauterine contraceptives (IUC) are highly effective methods with high continuation and satisfaction rates.^{1–3} However, fear of pain during insertion

can discourage some potential users, especially nulligravid women.^{4–6}

IUC insertion in nulligravid women is not more difficult than in parous women.⁷ However, nulligravidas report significantly more pain than parous women,^{8–11} which may be underestimated by healthcare providers (HCPs).¹² Although most women do not need any pain-reducing strategies for IUC insertion, some women may opt for these methods if they have received counseling about effective strategies to reduce the pain.

Several techniques have been tested to reduce pain during IUC insertion in nulligravidas. However, a majority of studies have found these techniques to be mostly ineffective, or the studies had small sample sizes.^{13–26} Local lidocaine injections appear to be effective in reducing pain during IUC insertion.^{15,16,27,28} In these studies, the lidocaine dose used ranged from 100 to 200 mg, administered through a paracervical block.

Unlike the paracervical block, which is a peripheral nerve block, the intracervical block acts as an infiltrative

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AJOG at a Glance

Why was this study conducted?

Paracervical blocks have been used to reduce pain at insertion of intrauterine contraceptives; however, intracervical block may be an easier and more reproducible technique than paracervical block. We conducted this study to examine whether a 3.6-mL 2% lidocaine intracervical block reduced pain during insertion of a 52-mg levonorgestrel-releasing intrauterine system (LNG-IUS) among nulligravid women.

Key findings

A 3.6-mL 2% lidocaine intracervical block decreased pain at tenaculum placement and LNG-IUS insertion in nulligravid women, and improved overall experience with the procedure.

What does this add to what is known?

Using a lower dose of lidocaine than reported with paracervical blocks, this study showed that a lidocaine intracervical block reduced pain at tenaculum placement and LNG-IUS insertion among nulligravidas.

anesthetic by distending the tissues, causing mechanical disruption of neural impulses. Theoretically, this requires a less precise injection than a nerve block and may be easier and more reproducible.²⁹ In light of the potential technical advantages of the intracervical block, our study evaluated whether a 3.6-mL 2% lidocaine intracervical block would reduce pain immediately after tenaculum placement and LNG-IUS insertion among nulligravid women. We also assessed whether the intracervical block would interfere with the ease of LNG-IUS insertion and with the overall experience with the LNG-IUS insertion.

Materials and Methods**Trial design and settings**

This was a randomized, double-blind, parallel, controlled trial conducted at 2 Brazilian university hospitals: the Family Planning division of the University Hospital of the Ribeirão Preto Medical School, University of São Paulo (HC-FMRP-USP), and the Family Planning Clinic of the Obstetrics and Gynecology Department at the University of Campinas (UNICAMP). The ethical review committees at each of these 2 institutions approved the study. All participants provided signed informed consent.

Participants were recruited between June 2017 and August 2018. The trial was

registered at [ClinicalTrials.gov](https://clinicaltrials.gov) (NCT 03111342).

Participants

Nulligravid women who desired to use the 52-mg LNG-IUS (20- μ g/d initial release) (Mirena, Bayer Oy, Turku, Finland) as a contraceptive, who were 18–45 years of age and who had never used any IUC, were included in the study. We excluded women with medical conditions considered category 3 or 4 for LNG-IUS use according to the World Health Organization (WHO) medical eligibility criteria³⁰ and women who had allergies or contraindications to lidocaine. Other exclusion criteria were a history of chronic pelvic pain, any abnormalities or history of surgery of the cervix, illicit drug or alcohol use, known psychiatric disorders, chronic use of medications that could interfere with pain perception (eg, antidepressants and anticonvulsants), and current use of analgesics or anti-inflammatory agents.

Consecutive sampling was used, as study personnel assessed all women who chose to use the LNG-IUS. The screening process lasted until the predetermined sample size was reached.

Interventions

Participants were randomized to 1 of 3 groups as follows: (1) intracervical

block: 3.6 mL 2% lidocaine (72 mg) without vasoconstricting agents, administered with a dental carpule syringe (2 tubes of 1.8 mL 2% lidocaine) and a 27-gauge needle in 4 points of the cervix (ie, 3, 6, 9, and 12 o'clock) prior to tenaculum placement ([Figure 1](#)); (2) sham: intracervical dry-needling, that is, insertion of the 27-gauge needle using the same syringe and technique as in the first group but without anesthetic; or (3) no intervention prior to tenaculum placement, which is the standard of care for IUC insertion.³¹

The rationale for using 72 mg lidocaine administered by an intracervical block came from a previous study conducted at our site.⁸ In that study, we compared an intracervical block with 36 mg of lidocaine to 400 mg of oral ibuprofen in women without prior vaginal deliveries undergoing a LNG-IUS insertion. As the pain at insertion was similar between the groups, we decided to double the lidocaine dose and to include only nulligravidas in this study.

We opted to use a dental syringe carpule because it has been used to administer intracervical anesthesia prior to cervical conization in our outpatient clinic. As the needle of this syringe is longer and thinner than usual, it seems to cause less pain and to be easier to use.

We chose intracervical dry-needling as the sham group because it uses the same intracervical block technique but without medication. In addition, dry-needling may reduce some types of pain, especially chronic pain.³²

The insertion was performed within the first 7 days of the menstrual cycle, or any time in case of consistent hormonal contraceptive use.

Outcomes and assessments

The primary outcome was pain measurement immediately after LNG-IUS insertion. Secondary outcomes were pain measurement immediately after tenaculum placement, ease of insertion, and overall experience with the LNG-IUS insertion procedure.

Pain was measured using 2 scales: a 10-cm visual analogue scale (VAS)³³ and a Faces Pain Scale from the International

Association for the Study of Pain (IASP).^{34,35} The Faces Pain Scale was a figure with faces with different expressions and corresponding numbers, ranging from 0 (no pain) to 5 (worst pain). Participants rated the pain using the VAS by touching a point on a line from 0 cm (no pain) to 10 cm (worst pain imaginable). They also pointed to the figure of the face whose expression best corresponded to their pain. One member of the team recorded the participants' selections in their file.

We evaluated the VAS score both as a categorical and a continuous variable. As a categorical variable, it was classified as no pain (0), mild pain (>0 to <4 cm), moderate pain (≥ 4 to <7 cm), or severe pain (≥ 7 to ≤ 10 cm). As a continuous variable, we considered mean pain score on the VAS and 95% confidence interval (95% CI).

We evaluated the Faces Pain Scale score as a categorical variable: no pain (0), mild pain (1), moderate pain (2 and 3), and severe pain (4 and 5).

The ease of LNG-IUS insertion was categorized by the HCP as usual, difficult, or very difficult shortly after the procedure.

To assess the participants' overall experience with LNG-IUS insertion, we asked them about pain and discomfort with insertion compared to their expectations, about their willingness to insert another LNG-IUS in the future, and whether they would recommend this procedure to others. We asked the pain and discomfort questions shortly after the insertion and after 24 hours (with a telephone call). For pain, we used a 5-point Likert scale ranging from "much less than expected" to "much more than expected." For discomfort, the scale ranged from "no discomfort" to "very uncomfortable." We asked the last 2 questions during a telephone call 24 hours after the procedure (and no later than 48 hours after the procedure). A nurse who was not involved in the procedure completed the phone calls.

We analyzed demographic and other variables including age, body mass index (BMI), race, household income, years of education, and prior use of HC. We also used anxiety and depression

FIGURE 1

Material used for intracervical block: dental carpule syringe, 2 tubes of 1.8 mL 2% lidocaine, and a 27-gauge needle



De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. *Am J Obstet Gynecol* 2020.

questionnaires to collect pre-procedure depression and anxiety scores.³⁶ Procedure-related variables were uterine position, uterine length measured with a uterine sound, need for cervical dilation, and the occurrence of any vagal reaction.

Randomization and blinding

Using a computer program (<https://www.sealedenvelope.com/simple-randomiser/v1/lists>), we randomized nulligravidas in blocks of 6 in a 1:1:1 allocation. One person who was not involved in the study performed the allocation sequence by placing the randomization numbers into sealed opaque envelopes. The randomization sequence was stored in a locked cabinet.

The study interventions were performed by 2 different HCPs. One HCP entered the examination room where the participant was awaiting the LNG-IUS insertion, opened the randomization envelope, completed the intervention to which the participant was allocated (intracervical block, intracervical sham, or no intervention), and placed the tenaculum. At this time, the participant

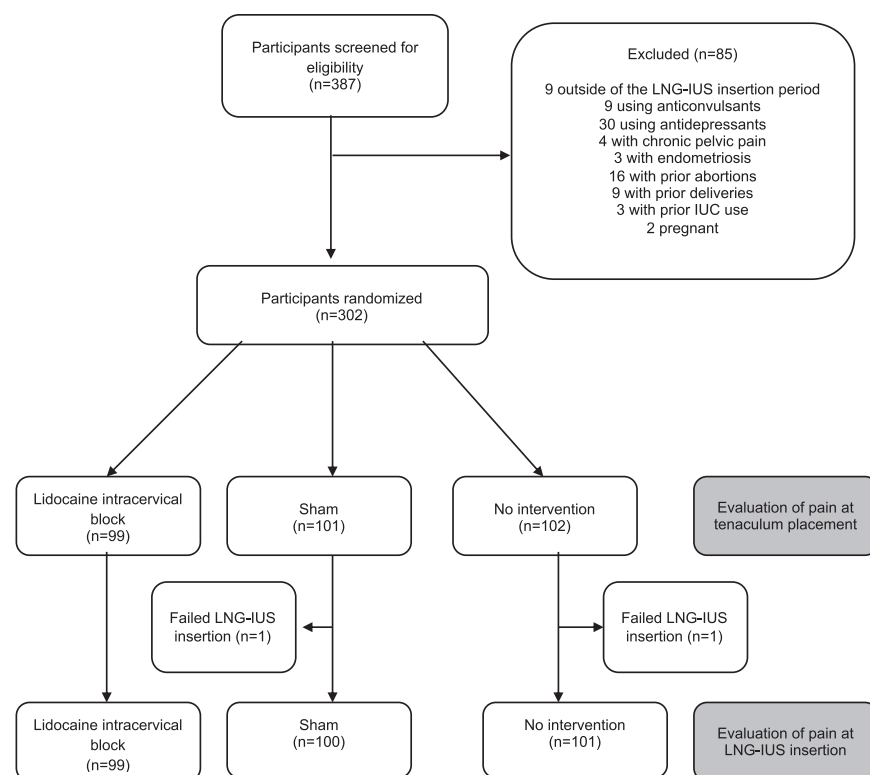
was asked to rate her pain with tenaculum placement. This HCP then left the room with the materials used for this procedure covered by a surgical sheet. The second HCP, who was blinded to the intervention performed as described above, then entered the examination room and inserted the LNG-IUS. Shortly after the procedure, the participant was asked to rate her pain with LNG-IUS insertion.

All HCPs involved were trained obstetrician-gynecologists, and no residents in training or mid-level providers were involved.

Sample size

Our sample size was powered to assess the pain at IUC insertion as measured by VAS score both as a continuous and a categorical variable.

In a previous study, nulliparous women undergoing IUC insertion without intervention for pain reduction reported a mean (standard deviation) VAS score of 5.9 (2.0) cm.²¹ To detect a 15% difference in VAS scores with an α value of 5% and 80% power, we calculated a sample size of 81 women per

FIGURE 2
Flow chart of the study

LNG-IUS: levonorgestrel-releasing intrauterine system; IUC: Intrauterine contraceptive

De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. *Am J Obstet Gynecol* 2020.

group. We selected a 15% change in pain score because reductions from 10% to 20% in the 10-cm VAS appear to reflect minimally important changes,³⁷ and we used the average of these for our sample size calculation.

A study conducted in 1 of the sites of this study showed that 56.5% of nulligravid women reported severe pain at LNG-IUS insertion.³⁸ We calculated a sample size of 97 women per group (291 in total) to detect an absolute difference of 20% in the proportion of women reporting severe pain at LNG-IUS insertion, with an α value of 5% and 80% power. We included 302 women to account for a 3.5% LNG-IUS insertion failure rate at the first attempt. This failure rate was based on the mean of the rates reported previously.^{39–41}

Statistical methods

We used the χ^2 test to compare categorical variables. For continuous

variables, we used analysis of variance (for baseline clinical, sociodemographic characteristics, and uterine length) and a linear mixed-effects regression model (for the VAS score).

To confirm how effective the blinding was, we asked the second HCP (who inserted the device) and the participant which group they thought they belonged to. We evaluated the blinding using the simple κ coefficient (95% CI).

We analyzed data for all participants, except for 2 who did not have the LNG-IUS inserted and did not rate the pain during insertion or overall experience with the procedure.

To evaluate the clinical significance (effect size)⁴² of the intracervical block for the pain-related outcomes, we calculated Cohen's d and number needed to treat (NNT — 95% CI to avert severe pain).⁴³ We used Cohen's d to compare mean pain scores as measured by the VAS scale.

We used SAS 9.4 software (SAS Institute, Cary, NC). The level of significance was set at 5%.

Results

We screened 387 nulligravidas, of whom 85 were excluded because they did not meet eligibility criteria, and 302 were randomized. A total of 99 participants were randomized to the intracervical block group; 101 were randomized to the sham group; and 102 to the no-intervention group (Figure 2). There were 2 failed LNG-IUS insertions (1 in the sham group and 1 in the no-intervention group). Clinical and demographic variables did not differ among the groups (Table 1).

Ease of LNG-IUS insertion and other variables related to LNG-IUS insertion did not differ among the groups. However, we did observe vagal reactions in 1.1%, 7.1%, and 6.9% of women in the intracervical block, sham, and no-intervention groups, respectively ($P = .08$) (Table 2). There were no uterine perforations.

Severe pain at tenaculum placement was less frequent in the intracervical group compared to the other groups (intracervical block: 2% vs sham: 30.2% vs no intervention: 15.2%, $P < .0001$) (Figure 3A). We found similar results with the Faces Pain Scale (Figure 4A). The mean (95% CI) pain score at tenaculum placement was lower in the intracervical block group than in the other groups (intracervical block: 2.2 [1.8–2.7] vs sham: 4.8 [4.4–5.3], $P < .0001$; intracervical block: 2.2 [1.8–2.7] vs no intervention: 4.2 [3.7–4.6], $P < .0001$). Pain at tenaculum placement did not differ between the sham and no-intervention groups ($P = .05$) (Figure 3B).

Severe pain at LNG-IUS insertion was less frequent in the intracervical block group than in the other groups (intracervical block: 26.5% vs sham: 59.4% vs no intervention: 50.5%, $P < .0001$) (Figure 3C). We found similar results with Faces Pain Scale (Figure 4B). The mean (95% CI) pain score at LNG-IUS insertion was lower in the intracervical block group than in the other groups (intracervical block:

4.3 [3.8–4.9] vs sham: 6.6 [6.2–7.0], $P < .0001$; intracervical block: 4.3 [3.8–4.9] vs no intervention: 5.8 [5.3–6.4], $P < .0001$. The group with no intervention had lower mean pain scores at LNG-IUS insertion than the sham group ($P = .02$) (Figure 3D).

The NNT (95% CI) for the intracervical block to avert severe pain at tenaculum placement and LNG-IUS insertion were 2 (1–6) and 4 (2–8), respectively. Cohen's d for mean pain (VAS) were -0.88 (large negative effect) and -0.54 (moderate negative effect) at tenaculum placement and LNG-IUS insertion, respectively.

Shortly after the procedure, 62.8%, 25%, and 36.7% of the participants in the intracervical block, sham, and no-intervention groups, respectively, rated their pain as lower than expected ($P < .0001$). More participants in the sham and no-intervention groups reported the LNG-IUS insertion as an uncomfortable or very uncomfortable procedure, compared to the intracervical block group ($P < .0001$). Only 1% of the women (3/300) who had a successful LNG-IUS insertion did not answer the questionnaire 24 hours after the procedure. We observed a similar result 24 hours after the procedure, with only 5.1% of participants in the intracervical block group reporting that they would not undergo the procedure again, compared to 18.2% in the sham group and 10% in the no-intervention group ($P = .01$). There was no difference among groups in the number of participants who reported that they would recommend the LNG-IUS insertion procedure to others (Table 3).

Blinding was considered satisfactory according to the κ coefficient, which showed no concordance between the actual group and that guessed by the HCP who inserted the LNG-IUS and by the participants (HCP: -10.3% [95% CI, -17.6% to 3.1%]; participants: -3.7% [95% CI, -11.8% to 4%]).

Comment

Principal findings

Our study showed that a 3.6-mL 2% lidocaine intracervical block decreased

TABLE 1

Baseline clinical and sociodemographic characteristics of nulligravid women

	Intracervical block n = 99	Sham n = 101	No intervention n = 102	
	Mean (SD) ^a	Mean (SD) ^a	Mean (SD) ^a	Pvalue
Age, y	25 (4.0)	24.7 (4.5)	25.1 (4.2)	.83
Monthly HHI, ^b USD	1963.2 (1563.2)	1764.1 (1715.0)	1940.2 (1649.5)	.67
BMI, kg/m ²	24.7 (5.1)	23.3 (4.9)	23.7 (3.9)	.11
Schooling, y	15.3 (2.5)	14.7 (2.5)	15.2 (2.3)	.19
ADS, ^b anxiety	5.1 (2.8)	5.2 (3.0)	5.1 (2.6)	.98
ADS, ^b depression	2.3 (2.0)	2.6 (2.0)	2.2 (2.0)	.40
	n (%) ^c	n (%) ^c	n (%) ^c	
Race				.45
White	90 (90.9)	84 (83.2)	86 (84.3)	
Mixed/black	6 (6.1)	12 (11.9)	13 (12.7)	
Asian	3 (3.0)	5 (4.9)	3 (2.4)	
HC use prior to randomization	94 (94.9)	93 (92.1)	100 (98.0)	.15

ADS, anxiety/depression screening score; BMI, body mass index; HC, hormonal contraceptives; HHI, household income; USD, US dollars.

^a Analysis of variance; ^b ADS score ≥ 7 is considered a positive screen for anxiety or depression.³⁶; ^c χ^2 test.

De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. Am J Obstet Gynecol 2020.

TABLE 2

Characteristics related to levonorgestrel-releasing intrauterine system insertion among nulligravid women

	Intracervical block n = 99	Sham n = 101	No intervention n = 102	
	n (%) ^a	n (%) ^a	n (%) ^a	Pvalue
Uterine position				.94
Anteverted	79 (84.9)	83 (86.4)	84 (86.6)	
Midline	7 (7.5)	6 (6.2)	8 (8.2)	
Retroverted	7 (7.5)	7 (7.2)	5 (5.1)	
Insertion				.35
Usual	93 (93.9)	88 (88.0)	85 (85.0)	
Difficult	5 (5.0)	11 (11.0)	13 (13.0)	
Very difficult	1 (1.0)	1 (2.0)	2 (2.0)	
Cervical dilation ^b	5 (5.1)	9 (9.1)	12 (12.0)	.23
Vagal reaction	1 (1.1)	7 (7.1)	7 (6.9)	.08
	Mean (SD) ^c	Mean (SD) ^c	Mean (SD) ^c	
Uterine length ^d (cm)	7.3 (0.6)	7.1 (0.6)	7.2 (0.5)	.29

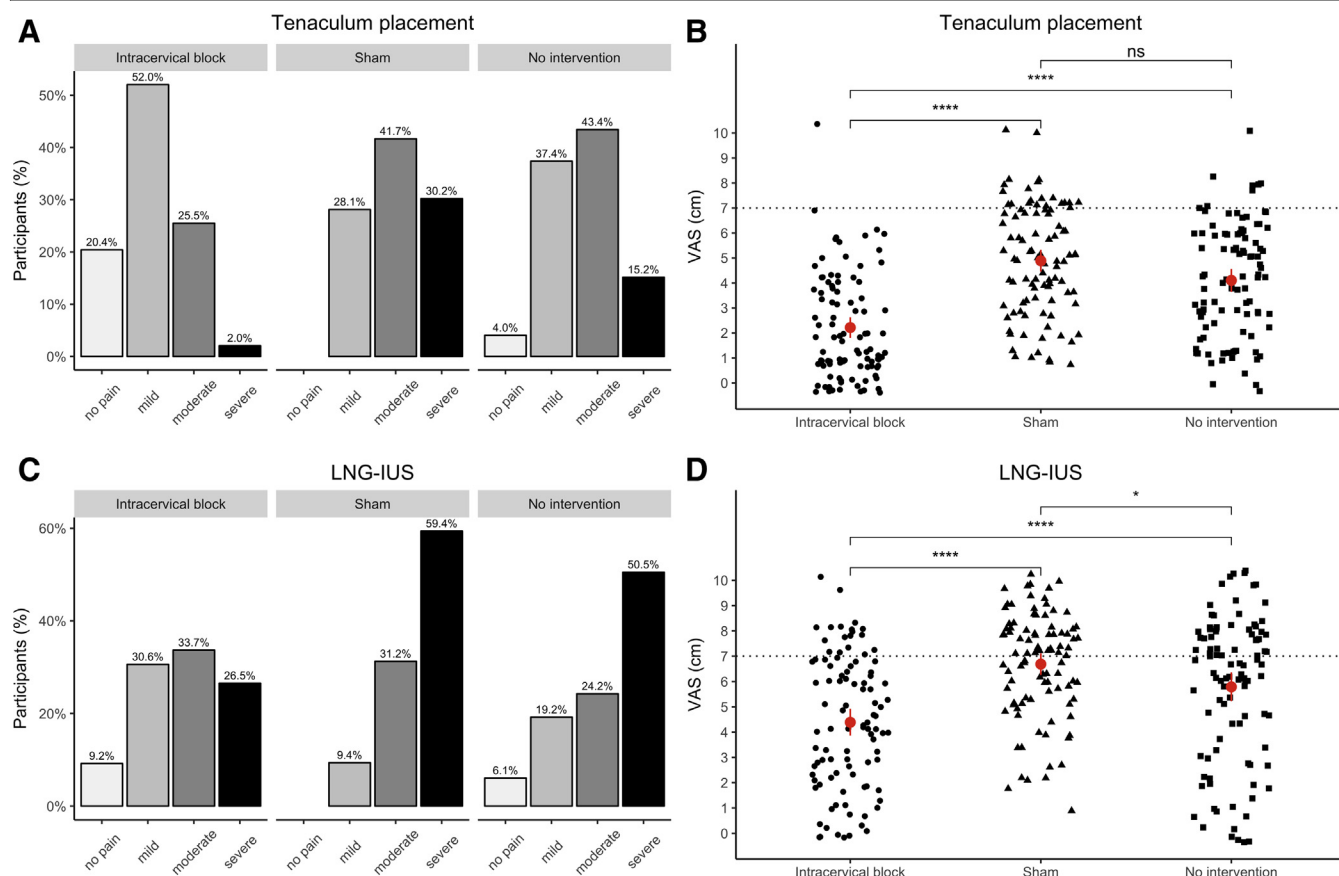
SD, standard deviation.

^a χ^2 test; ^b Dilated using Os Finder Cervical Dilator; ^c Analysis of variance; ^d Measured with a uterine sound.

De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. Am J Obstet Gynecol 2020.

FIGURE 3

Pain related to the LNG-IUS insertion procedure among nulligravid women. A and B, Pain at tenaculum placement. C and D, Pain at LNG-IUS insertion. Pain was measured using the VAS. LNG-IUS, levonorgestrel-releasing intrauterine system; NS, not statistically significant; VAS, visual analogue scale. * $P = .02$; **** $P < .0001$. A and C, $P < .0001$. B and D, Red dot is the mean pain score, and red vertical bar is the 95% confidence interval



De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. Am J Obstet Gynecol 2020.

pain immediately after tenaculum placement and with LNG-IUS insertion in nulligravidas compared to an intracervical sham injection and to no intervention. We also found that fewer participants in the intracervical block group reported severe pain immediately after tenaculum placement and LNG-IUS insertion than in the other groups. Women who received an intracervical block also reported an overall more positive experience with LNG-IUS insertion than those in the other groups: they had lower levels of discomfort, had less pain than they had expected, and were more likely to be willing to insert a new LNG-IUS in the future. Despite of these findings, the intracervical block had no positive

effect on provider ease of LNG-IUS insertion.

We evaluated the clinical significance of our pain-related results by calculating the NNT and Cohen's d . The NNT for the intracervical block to avert severe pain at tenaculum placement was 2, and at LNG-IUS insertion it was 4. Single-digit values for NNT usually denote a worthwhile difference when comparing one intervention to another.⁴² For the mean pain score as measured by the VAS, the effect size for the intracervical block was considered to be large; for the LNG-IUS insertion it was moderate.

Results in context of what is known

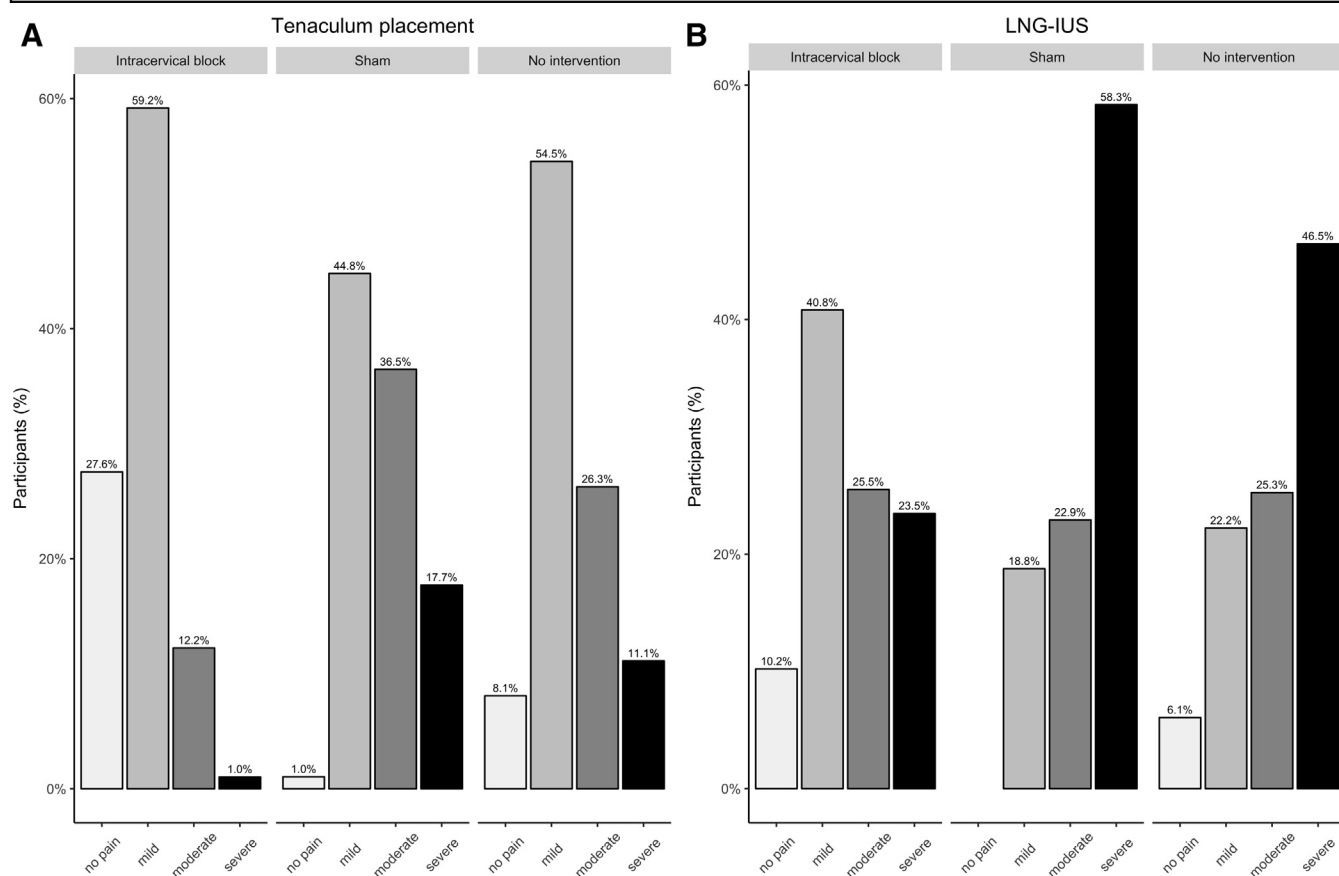
Few studies have investigated the use of lidocaine for pain reduction at IUC

insertion among nulligravidas.^{13–16} Use of 2% lidocaine gel has been shown to reduce pain at tenaculum placement, but did not have any effect on IUC insertion.¹³ Another study using a novel topical lidocaine formulation (short-acting 4% viscous solution) not available on the market showed reduced pain at IUC insertion.¹⁴ Two studies of paracervical block at IUC insertion showed pain reduction in nulligravidas using higher doses of lidocaine (100 and 200 mg) than in our study (72 mg).^{15,16}

In 1 study, intracervical block with 36 mg of lidocaine was compared to 400 mg of oral ibuprofen in women without prior vaginal deliveries undergoing LNG-IUS insertion. There was no difference among groups in terms of pain at

FIGURE 4

Pain related to the LNG-IUS insertion procedure among nulligravid women. A, Pain at tenaculum placement. B, Pain at LNG-IUS insertion. Pain was measured using the Faces Pain Scale
LNG-IUS, levonorgestrel-releasing intrauterine system. A and B, $P < .0001$.



De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. *Am J Obstet Gynecol* 2020.

insertion.⁸ A multivariate analysis showed that nulliparity was associated with an almost 3-fold increase in risk of severe pain compared to having had a prior cesarean delivery. In that study, the dose of lidocaine was low, and there was no subanalysis of nulligravid women.

Our study did not show any benefit of the intracervical block for the ease of LNG-IUS insertion. This is consistent with prior studies using anesthetics at IUC insertion.⁴⁴ In our study, less than 15% of insertions were considered to be difficult or very difficult across groups; this is a low rate that does not justify the routine use of interventions to facilitate insertion. It is not known whether anesthetic blocks would confer benefit during difficult insertions.

Clinical implications

Despite the fact that the use of 3.6-mL 2% intracervical lidocaine significantly reduced severe pain at LNG-IUS insertion and improved the procedure experience among nulligravidas, almost half of the participants with no intervention did not report severe pain, and therefore would be unlikely to need any intervention for pain control at LNG-IUS insertion. Therefore, even among nulligravidas, who are at higher risk for experiencing severe pain,⁸ the intracervical block should not be routinely performed at IUC insertion. Instead, it can be offered as an option to reduce pain and discomfort associated with the procedure.

Research implications

Future studies exploring whether an intracervical block decreases pain among other groups of women at risk for higher pain scores⁴⁵ (eg, women with previous cesarean delivery, women with difficult IUC insertion) are necessary to provide evidence-based clinical guidance for pain management at IUC insertion.

Women from the sham and the no-intervention groups had 7 times more vagal episodes than those from the intracervical block group. Although this comparison did not reach statistical significance because of the small numbers of episodes, this finding may have clinical significance and should be addressed in future studies.

TABLE 3

Overall experience with levonorgestrel-releasing intrauterine system insertion for nulligravid women randomized to intracervical block, sham, and no intervention

	Intracervical block n = 99	Sham n = 100	No Intervention n = 101	
	Immediately after LNG-IUS insertion			
	n (%)	n (%)	n (%)	Pvalue ^a
Pain experienced compared to expected				<.0001
Much less	24 (24.7)	2 (2.0)	13 (12.9)	
Less	37 (38.1)	23 (23.0)	24 (23.8)	
Same	21 (21.6)	28 (28.0)	19 (18.8)	
More	12 (12.4)	39 (39.0)	31 (30.7)	
Much more	3 (3.1)	8 (8.0)	14 (13.9)	
Discomfort with insertion				<.0001
Very uncomfortable	11 (11.1)	29 (29.3)	30 (29.7)	
Uncomfortable	41 (41.4)	52 (52.5)	50 (49.5)	
Somewhat uncomfortable	44 (44.4)	15 (15.1)	18 (17.8)	
Comfortable	3 (3.0)	3 (3.0)	3 (3.0)	
	24 h after LNG-IUS insertion			
	n (%)	n (%)	n (%)	
Would insert again				.01
Yes	93 (94.9)	81 (81.8)	90 (90.0)	
No	5 (5.1)	18 (18.2)	10 (10.0)	
Pain experienced compared to expected				<.0001
Much less	19 (19.4)	2 (2.0)	7 (7.0)	
Less	38 (38.8)	17 (17.2)	22 (22.0)	
Same	18 (18.4)	18 (18.2)	15 (15.0)	
More	17 (17.3)	37 (37.4)	33 (33.0)	
Much more	6 (6.1)	25 (25.2)	23 (23.2)	
Discomfort with insertion				<.0001
Very uncomfortable	11 (11.2)	35 (35.3)	26 (26.0)	
Uncomfortable	49 (50.0)	56 (56.6)	52 (52.0)	
Somewhat uncomfortable	37 (37.8)	6 (6.1)	20 (20.0)	
Comfortable	1 (1.0)	2 (2.0)	2 (2.0)	
Would recommend to others				.22
Yes	97 (99.0)	93 (94.9)	94 (94.9)	
No	1 (1.0)	5 (5.1)	5 (5.0)	

^a χ^2 test.

De Nadai et al. Intracervical block for LNG-IUS insertion among nulligravid women. Am J Obstet Gynecol 2020.

Strengths and limitations

Limitations of this study included the lack of diversity in terms of participants' race, income, and schooling. Also, participants had higher educational attainment and income compared to the

general population attending public hospitals in Brazil. However, the cities where the study was conducted are wealthier and have a higher human development index than the national average.⁴⁶ Also, although an intracervical

block can produce some pain, we did not evaluate the pain with this intervention, and opted to capture this information indirectly by assessing the overall experience with LNG-IUS insertion. Another limitation is that we did not include

adolescents less than 18 years of age. However, age does not seem to affect pain at IUC insertion, whereas nulliparity does.^{9,10} Because we excluded those individuals less than 18 years or with any prior pregnancies, we automatically selected for women with higher educational attainment and socioeconomic status, as well as for fewer black women.⁴⁷ Finally, we excluded some conditions that could interfere with pain perception, such as chronic pelvic pain and known psychiatric disorders. These exclusions and the lack of diversity may compromise the generalizability of our results.

Our study does have strengths. It is the largest study to assess the use of local injectable lidocaine for IUC insertion among nulligravid women, a group at high risk for severe pain at IUC insertion.^{8–11} It is a double-blind randomized controlled trial, which reduces the chance of detection and performance bias. In Brazil, the rate of functional illiteracy is 23%, which can affect participants' ability to understand and correctly fill out the VAS.^{48,49} We added the Faces Pain Scale to measure pain to ensure that our results were reliable independent of the scale used. In addition, we evaluated the intracervical block, which may be easier to perform than a paracervical block²⁹; we used a lower dose of lidocaine than previous studies of paracervical blocks^{15,16}; and we inserted only 1 type of IUC.

Conclusion

One of the main barriers to acceptability of IUCs is fear of pain at insertion.^{4,50} In this context, it is important to inform potential users that methods exist to reduce this pain. Use of a 3.6-mL 2% lidocaine intracervical block is a strategy that can be offered to reduce severe pain and discomfort associated with LNG-IUS insertion among nulligravid women. ■

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